Rossmoyne Senior High School

Semester One Examination, 2023

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

If required by your examination administrator, please place your student identification label in this box

WA student number: In figures

In words

Circle your Teacher's Name: Mr Fletcher Mrs Fraser-Jones Mrs Murray Mr Pisano Mr Stephens

Time allowed for this section

Reading time before commencing work:	five minutes
Working time:	fifty minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen.
Do not use erasable or gel pens.

3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
1	7	
2	6	
3	9	
4	9	
5	8	
6	6	
7	7	
S1 Total	52	
S1 Wt (×0.6731)	35%	
S2 Wt	65%	
Total	100%	

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Section One: Calculator-free **35% (52 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1 **(7 marks)**

Consider the matrices $\boldsymbol{P} = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$, $\boldsymbol{Q} = \begin{bmatrix} 3 & 0 \end{bmatrix}$, $\boldsymbol{R} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$ and $\boldsymbol{S} = \begin{bmatrix} 4 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$.

(a) The size of the matrix that has a zero in its second row and first column is $a \times b$. State the value of a and the value of b. (1 mark)

(b) Calculate $\boldsymbol{P} + 4\boldsymbol{I}$, where $\boldsymbol{I}$ is the 2×2 identity matrix. (2 marks)

(c) The sum of matrices $\boldsymbol{S}$ and $\boldsymbol{T}$ is $\begin{bmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$. Determine matrix $\boldsymbol{T}$. (2 marks)

(d) Calculate the matrix product $\boldsymbol{PS}$. (2 marks)

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Question 2 **(6 marks)**

(a) Given $\frac{ab}{c} + d = e$,

(i) find d if $a = -1$, $b = 4$, $c = 1$ and $e = -10$. (2 marks)

(ii) find c if $a = -2$, $b = 3$, $d = -4$ and $e = -2$. (2 marks)

(b) Simplify the expression $\frac{14 + 2x}{x^2 + x - 8}$ when $x = -5$. (2 marks)

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Question 3 **(9 marks)**

The diagram at right, shows a model of a right pyramid with its apex 12 cm directly above the centre of its square base.

The length of one side of the square is 10 cm and the distance from the apex to the midpoint of one side of the square is 13 cm.

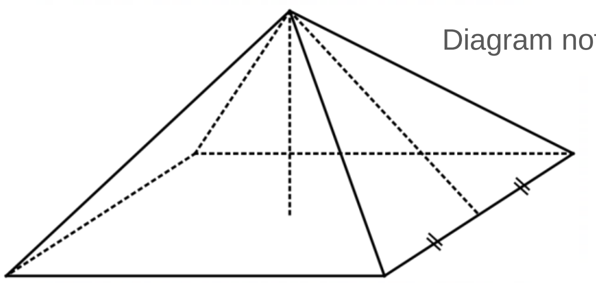

Diagram not to scale.

(a) Clearly label the diagram with the three given dimensions. (1 mark)

(b) Determine the volume of the pyramid. (2 marks)

(c) Determine the total surface area of the pyramid. (3 marks)

(d) A playground pyramid is to be constructed at the local park, based on this model pyramid. The scale ratio of lengths of the model pyramid compared to the playground pyramid is 1:50. Determine the number of square metres of land that the playground pyramid will cover.

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Question 4 **(9 marks)**

The table below shows the buy and sell rates for one Australian dollar advertised by an airport foreign exchange kiosk for several currencies.

Currency (country)	Buy rate	Sell rate
Pound (United Kingdom)	0.6	0.55
Kroner (Swedish)	7.00	6.50
Franc (New Caledonia)	80	72

(a) Explain why the kiosk would use the sell rate if a person wanted to exchange $\$200$ for francs and state how many francs the person would receive. (2 marks)

(b) A traveller returned from the United Kingdom with 60 pounds. Determine how many dollars they would receive in exchange from the kiosk. (2 mark)

(c) The kiosk sets the sell rate for the New Caledonian francs at 90% of the value of the buy rate.
If the current buy rate decreased by 5 francs, determine the new sell rate. (2 marks)

(d) A student exchanged $\$400$ into Swedish kroner at the kiosk. They spent 2320 kroner during their trip to Sweden and exchanged the remaining kroner into dollars at the kiosk on their return to Australia. Determine how many dollars they received. (3 marks)

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Question 5 **(8 marks)**

Dylan has created the spreadsheet shown to model his weekly spending budget.

	A	B
1	Rent	\$180
2	Electricity	
3	Transport	\$55
4	Food	\$210
5	Other	\$130
6	Total	\$600

(a) Determine the amount that Dylan has budgeted for electricity. (2 mark)

Dylan currently works 4 days per week at a vineyard where he is paid $\$6.50$ cash-in-hand for each box of grapes that he picks.

(b) If Dylan picks 30 boxes of grapes per day, determine how much of his earnings remain each week after he has spent all his budget and put aside 10% of his earnings for tax. (3 marks)

(c) Dylan has been told that his rent is going to increase by 15% but plans to reduce his ‘other’ spending of $\$130$ by 20%. Determine the overall effect that these two changes will have on his weekly spending budget. (3 marks)

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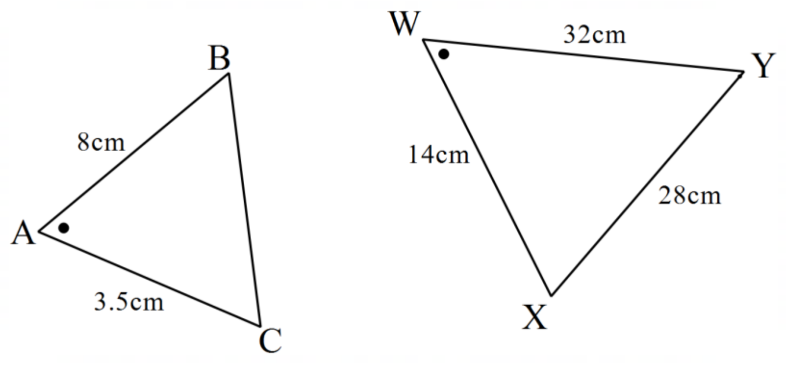
Question 6 **(6 marks)**

These two triangles are similar. Note that they are not drawn to scale.



(a) Write a similarity statement for the triangles above. (2 marks)

(b) Determine the length of the missing side. (1 mark)

(c) The area of ΔWXY is 192cm^2. Use the length scale factor to calculate the area of ΔABC. (2 marks)

(d) These triangles are the front view of two similar triangular prisms.
What is the ratio of their volumes? (1 mark)

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Question 7 **(7 marks)**

(a) The gear ratio R of a mechanical system is calculated using $R = \frac{d_2}{d_1 + 8} - \frac{d_1}{d_2 - 1}$, where d_1 and d_2 are the diameters of two cogs.

Determine the value of R when $d_1 = 4$ cm and $d_2 = 9$ cm. (2 marks)

(b) When an object is placed at a distance of x cm from a lens with focal length of f cm, its image will appear at a distance d cm from the lens, where $d = \frac{xf}{x - f}$. An object is 12.5 cm from a lens that has a focal length of 10 cm. Determine how far its image will appear from the lens. (2 marks)

(c) The velocity of a model car is given by $v = \sqrt{u^2 + 2as}$ cm/s. Determine

(i) the velocity of the model car when $u = 4, a = 1.5$ and $s = 3$. (1 mark)

(ii) the change in velocity of the model car when $u = 4$ and $a = 1.5$, as the value of s increases from 3 to 16. (2 marks)

End of questions

Supplementary page

Question number: _________

Supplementary page

Question number: _________